

Your Paper

You

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Abstract

Your abstract.

1 PID discreto

A equação do PID discreto pode ser escrita como:

$$C(z) = P(z) + I(z) + D(z) = \frac{y(z)}{e(z)} = \frac{y_p(z) + y_i(z) + y_d(z)}{e(z)} = K_p + K_i \frac{T_s}{2} \frac{z+1}{z-1} + K_d \frac{N}{1 + NT_s \frac{1}{z-1}} \quad (1)$$

Discretizaremos ela em partes, começando de:

2 Termo proporcional

$$P(z) = \frac{y_p(z)}{e(z)} = K_p \quad (2)$$

$$P(z) = y_p(z) = K_p e(z) \quad (3)$$

No tempo:

$$\mathcal{Z}^{-1} [P(z)] = \mathcal{Z}^{-1} [y_p(z)] = \mathcal{Z}^{-1} [K_p e(z)] \quad (4)$$

$$y_p[k] = K_p e[k] \quad (5)$$

3 Termo integrativo

$$I(z) = \frac{y_i(z)}{e(z)} = K_i \frac{T_s}{2} \frac{z+1}{z-1} \quad (6)$$

$$\frac{y_i(z)}{e(z)} = K_i \frac{T_s}{2} \frac{z+1}{z-1} \quad (7)$$

$$y_i(z) = e(z) \left[K_i \frac{T_s}{2} \frac{z+1}{z-1} \right] \quad (8)$$

$$y_i(z) = \frac{K_i T_s}{2} e(z) \left[\frac{z+1}{z-1} \right] \times \frac{z^{-1}}{z^{-1}} \quad (9)$$

$$y_i(z) = \frac{K_i T_s}{2} e(z) \left[\frac{1+z^{-1}}{1-z^{-1}} \right] \quad (10)$$

$$y_i(z) = \frac{K_i T_s}{2} \left[\frac{e(z) + z^{-1} e(z)}{1 - z^{-1}} \right] \quad (11)$$

$$y_i(z) [1 - z^{-1}] = \frac{K_i T_s}{2} [e(z) + z^{-1}e(z)] \quad (12)$$

$$[y_i(z) - y_i(z)z^{-1}] = \frac{K_i T_s}{2} [e(z) + z^{-1}e(z)] \quad (13)$$

No tempo:

$$\mathcal{Z}^{-1} [y_i(z)] - \mathcal{Z}^{-1} [y_i(z)z^{-1}] = \frac{K_i T_s}{2} [\mathcal{Z}^{-1} [e(z)] + \mathcal{Z}^{-1} [z^{-1}e(z)]] \quad (14)$$

$$y_i(k) - y_i(k-1) = \frac{K_i T_s}{2} [e(k) + e(k-1)] \quad (15)$$

$$y_i(k) = \frac{K_i T_s}{2} [e(k) + e(k-1)] + y_i(k-1) \quad (16)$$

4 Termo derivativo

$$D(z) = \frac{y_d(z)}{e(z)} = K_d \frac{N}{1 + NT_s \frac{1}{z-1}} \quad (17)$$

$$\frac{y_d(z)}{e(z)} = K_d \frac{N}{1 + NT_s \frac{1}{z-1}} \quad (18)$$

$$y_d(z) = K_d N \frac{e(z)}{1 + NT_s \frac{1}{z-1}} \quad (19)$$

$$y_d(z) \left[1 + NT_s \frac{1}{z-1} \right] = K_d N e(z) \quad (20)$$

$$y_d(z) [z - 1 + NT_s] \times \frac{z^{-1}}{z-1} = K_d N e(z) [z - 1] \times \frac{z^{-1}}{z-1} \quad (21)$$

$$y_d(z) [1 - (1 - NT_s)z^{-1}] = K_d N e(z)(1 - z^{-1}) \quad (22)$$

$$y_d(z) - (1 - NT_s)y_d(z)z^{-1} = K_d N [e(z) - e(z)z^{-1}] \quad (23)$$

No tempo:

$$\mathcal{Z}^{-1} [y_d(z)] - (1 - NT_s)\mathcal{Z}^{-1} [y_d(z)z^{-1}] = K_d N [\mathcal{Z}^{-1} [e(z)] - \mathcal{Z}^{-1} [e(z)z^{-1}]] \quad (24)$$

$$y_d(k) - (1 - NT_s)y_d(k-1) = K_d N [e(k) - e(k-1)] \quad (25)$$

$$y_d(k) = K_d N [e(k) - e(k-1)] + (1 - NT_s)y_d(k-1) \quad (26)$$

5 PID

A saída do controlador pode ser expressa como:

$$y(k) = y_p(k) + y_i(k) + y_d(k) \quad (27)$$